

# MTH312 - HW2

## Group 5

2026-02-16

## Question: 1

### Problem Definition

The objective of this analysis is to study the relationship between **Weekly Sales** and a set of explanatory variables — and **Temperature, Fuel, Price, Consumer Price Index (CPI) and Unemployment** using **Linear Quantile Regression**.

Classical ordinary linear regression estimates the conditional mean  $E[Y|X]$  (assuming a constant variance), and is optimal under a squared-error loss. However, sales data are often characterized by **skewness, heavy tails and extreme observations**, which can substantially influence mean-based estimates.

To overcome these limitations, quantile regression is employed to model the **entire conditional distribution** of weekly sales. By estimating multiple conditional quantiles (e.g.  $\tau = 0.05, 0.25, 0.50, 0.75, 0.95$ ) we can examine how the effects of covariates differ across low, median and high-sales regimes.

## Methodology

### Ordinary Least Squares Regression

OLS regression estimates the conditional mean by solving

$$\hat{\beta}_{OLS} = \arg \min_{\beta} \sum_{i=1}^n (y_i - x_i^T \beta)^2$$

While efficient under homoscedastic normal errors, OLS is sensitive to outliers and provides limited insight into distributional heterogeneity.

### Linear Quantile Regression

Quantile regression, introduced by **Koenker and Bassett (1978)**, extends linear regression by estimating conditional quantiles of the response variable. For a given quantile level, the conditional quantile function is defined as

$$Q_Y(\tau | X) = X^T \beta(\tau).$$

The quantile regression estimator is obtained by minimizing the check (pinball) loss function:

$$\hat{\beta}(\tau) = \arg \min_{\beta} \sum_{i=1}^n \rho_{\tau}(y_i - x_i^{\top} \beta),$$

where

$$\rho_{\tau}(u) = u(\tau - \mathbb{I}(u < 0)).$$

This formulation allows the impact of explanatory variables to vary across different parts of the conditional distribution.

# R Implementation

## Setup and Data Loading

The analysis uses a panel dataset of weekly retail sales observations combined with contemporaneous macroeconomic and weather variables. Weekly sales constitute the outcome variable, while temperature captures local weather conditions, fuel price proxies transportation and energy costs, CPI represents inflation and unemployment reflects labor market conditions. The dataset exhibits substantial dispersion and clustering across covariates, motivating the use of quantile regression to examine heterogeneous effects across the sales distribution.

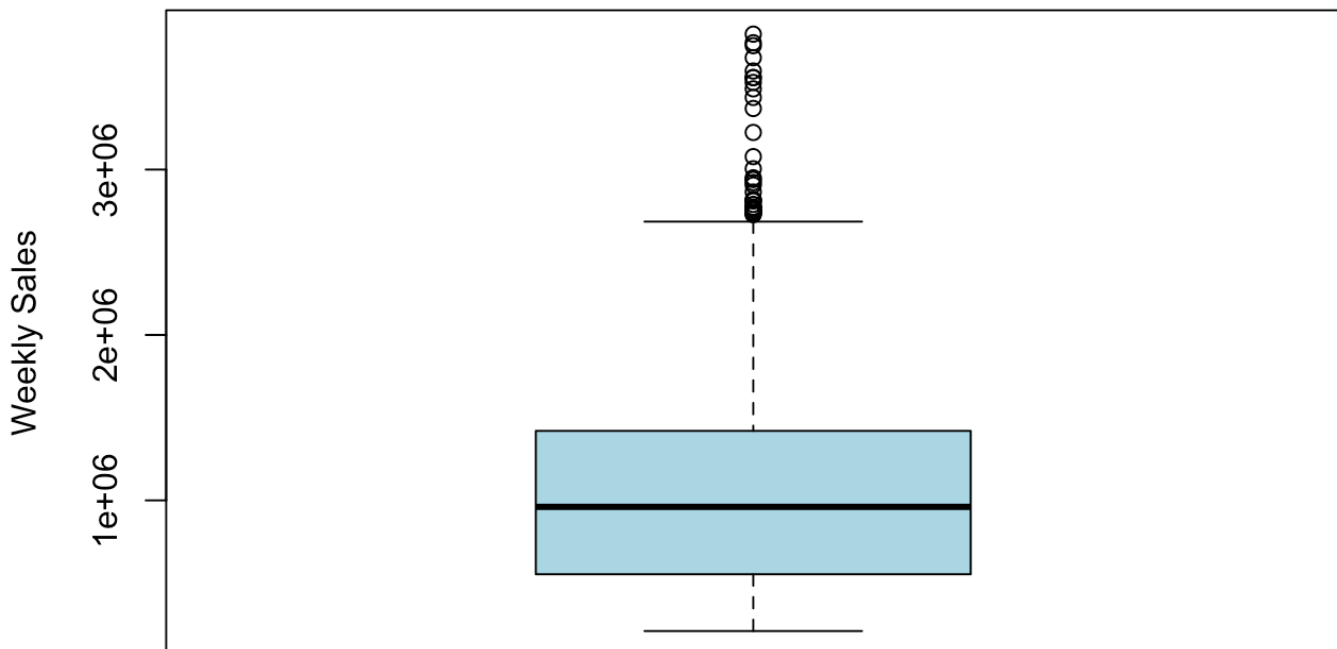
```
set.seed(42)

#load dataset
walmart <- read.csv("Walmart_Store_sales.csv")
summary(walmart)
```

```
##      Store      Date      Weekly_Sales      Holiday_Flag
## Min.      : 1      Length:6435      Min.      : 209986      Min.      :0.00000
## 1st Qu.:12      Class :character      1st Qu.: 553350      1st Qu.:0.00000
## Median :23      Mode  :character      Median : 960746      Median :0.00000
## Mean   :23                      Mean   :1046965      Mean   :0.06993
## 3rd Qu.:34                      3rd Qu.:1420159      3rd Qu.:0.00000
## Max.   :45                      Max.   :3818686      Max.   :1.00000
## Temperature      Fuel_Price      CPI      Unemployment
## Min.      : -2.06      Min.      :2.472      Min.      :126.1      Min.      : 3.879
## 1st Qu.: 47.46      1st Qu.:2.933      1st Qu.:131.7      1st Qu.: 6.891
## Median : 62.67      Median :3.445      Median :182.6      Median : 7.874
## Mean   : 60.66      Mean   :3.359      Mean   :171.6      Mean   : 7.999
## 3rd Qu.: 74.94      3rd Qu.:3.735      3rd Qu.:212.7      3rd Qu.: 8.622
## Max.   :100.14      Max.   :4.468      Max.   :227.2      Max.   :14.313
```

```
#checking for outliers
boxplot(walmart$Weekly_Sales, main = "Boxplot of Weekly Sales", ylab = "Weekly Sales", col = "lightblue")
```

### Boxplot of Weekly Sales



The boxplot of weekly sales reveals a strongly **right-skewed distribution with numerous extreme upper outliers**. These observations correspond to unusually high sales weeks and indicate substantial heterogeneity in the variability of sales across time. The presence of **skewness, heavy tails, and heteroskedasticity** suggests that mean regression alone may provide an incomplete summary of the conditional distribution. This motivates the use of quantile regression, which allows the effects of explanatory variables to be examined across different parts of the sales distribution and is robust to extreme observations.

## Model Implementation

Quantile regression models were estimated for

$$\tau = 0.05, 0.25, 0.5, 0.75, 0.95$$

using the `quantreg` package in R. OLS regression was used as a baseline for comparison. Coefficient estimates across quantiles were compared to assess how the effects of temperature, fuel price, CPI, and unemployment vary across the sales distribution.

```
library(quantreg)
library(ggplot2)
library(dplyr)
library(plotly)
#OLS(baseline)
ols_model <- lm(Weekly_Sales ~ Temperature + Fuel_Price + CPI + Unemployment, data
= walmart)
summary(ols_model)
```

```
##
## Call:
## lm(formula = Weekly_Sales ~ Temperature + Fuel_Price + CPI +
##     Unemployment, data = walmart)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -956788 -478860 -115969  394736 2789686
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1743607.6    79553.1   21.918 < 2e-16 ***
## Temperature    -885.7      396.2   -2.235  0.0254 *
## Fuel_Price    -12248.4    15751.8   -0.778  0.4368
## CPI           -1585.8      195.2   -8.126  5.3e-16 ***
## Unemployment  -41215.0     3972.7 -10.375 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 557600 on 6430 degrees of freedom
## Multiple R-squared:  0.02433,    Adjusted R-squared:  0.02372
## F-statistic: 40.09 on 4 and 6430 DF,  p-value: < 2.2e-16
```

```
#quantile regression models
taus <- c(0.05, 0.25, 0.5, 0.75, 0.95)

quantile_regression_models <- lapply(taus, function(tau) {
  rq(Weekly_Sales ~ Temperature + Fuel_Price + CPI + Unemployment, tau = tau, data
= walmart)
})

#comparing coefficients across different regression quantiles
coef_table <- sapply(quantile_regression_models, coef)
colnames(coef_table) <- paste0("tau=", taus)
round(coef_table, 2)
```

```
##          tau=0.05   tau=0.25   tau=0.5   tau=0.75   tau=0.95
## (Intercept) 75416.89 1430873.10 1720742.51 2276871.05 3098217.36
## Temperature -1872.68   -2346.18    290.84    3901.85   -4269.25
## Fuel_Price  31514.48   -8007.91   1335.85   -21606.35  -73392.18
## CPI          963.18    -2076.25   -3341.30   -2284.20    -30.47
## Unemployment 11636.07  -41056.37  -32990.98  -75971.21  -71391.36
```

```
#calculating mse
compute_mse <- function(y_true, y_pred) {
  mean((y_true - y_pred)^2)
}

#mse for ols
ols_pred <- predict(ols_model, newdata = walmart)
mse_ols <- compute_mse(walmart$Weekly_Sales, ols_pred)
mse_ols
```

```
## [1] 310711823081
```

```
#mse for quantile regressions
mse_quantiles <- sapply(seq_along(taus), function(i) {

  qr_pred <- predict(quantile_regression_models[[i]],
                    newdata = walmart)

  compute_mse(walmart$Weekly_Sales, qr_pred)
})

#comparison of mse
mse_comparison <- data.frame(
  Model = c("OLS", rep("Quantile Regression", length(taus))),
  Quantile = c(NA, taus),
  MSE = c(mse_ols, mse_quantiles)
)
mse_comparison
```

| Model<br><chr>      | Quantile<br><dbl> | MSE<br><dbl> |
|---------------------|-------------------|--------------|
| OLS                 | NA                | 3.107118e+11 |
| Quantile Regression | 0.05              | 8.431822e+11 |
| Quantile Regression | 0.25              | 5.328696e+11 |
| Quantile Regression | 0.50              | 3.365547e+11 |
| Quantile Regression | 0.75              | 4.758205e+11 |

| Quantile Regression | 0.95 | 1.264656e+12 |
|---------------------|------|--------------|
| 6 rows              |      |              |

OLS attains the lowest mean squared error, reflecting its optimality for predicting the conditional mean under squared-error loss. The median quantile ( $\tau = 0.50$ ) performs closest to OLS, while extreme quantiles have higher MSE because they target tail behavior rather than mean accuracy. Thus, MSE favors OLS but does not assess the distributional insights provided by quantile regression.

```

plot_quantile_2d <- function(var_name) {

  f_qr <- as.formula(paste("Weekly_Sales ~", var_name))
  f_ols <- as.formula(paste("Weekly_Sales ~", var_name))

  qr_model <- rq(f_qr, tau = taus, data = walmart)
  ols_model <- lm(f_ols, data = walmart)

  coef_mat <- coef(qr_model)

  p <- ggplot(walmart, aes_string(x = var_name, y = "Weekly_Sales")) +
    geom_point(alpha = 0.4) +
    geom_abline(
      intercept = coef(ols_model)[1],
      slope = coef(ols_model)[2],
      color = "red",
      linewidth = 1.2
    )

  # add quantile lines
  for (i in seq_along(taus)) {
    p <- p + geom_abline(
      intercept = coef_mat[1, i],
      slope = coef_mat[2, i],
      linetype = "dashed",
      color = scales::hue_pal()(length(taus))[i],
      linewidth = 1
    )
  }

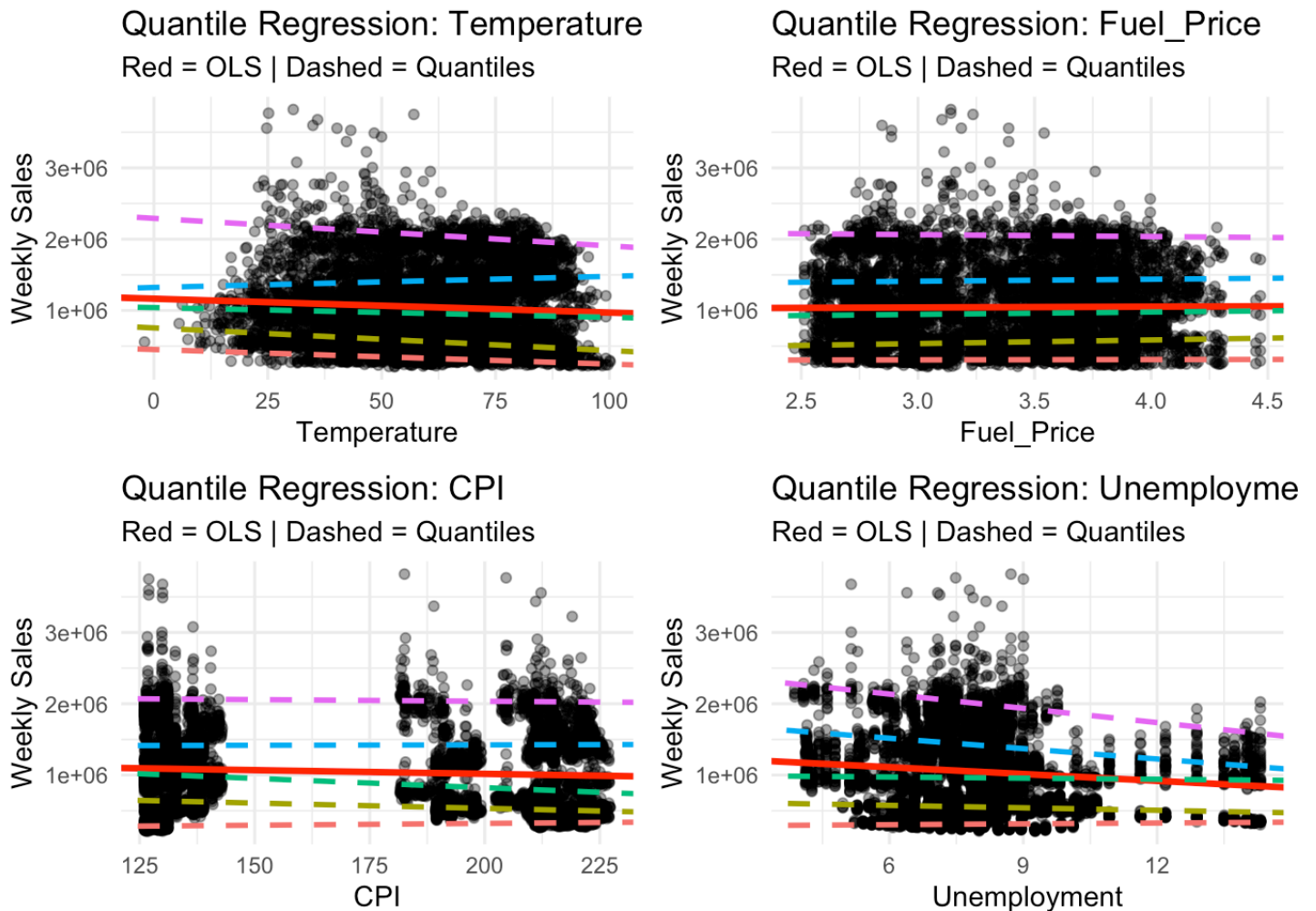
  p +
    labs(
      title = paste("Quantile Regression:", var_name),
      subtitle = "Red = OLS | Dashed = Quantiles",
      x = var_name,
      y = "Weekly Sales"
    ) +
    theme_minimal()
}

library(gridExtra)

p1 <- plot_quantile_2d("Temperature")
p2 <- plot_quantile_2d("Fuel_Price")
p3 <- plot_quantile_2d("CPI")
p4 <- plot_quantile_2d("Unemployment")

grid.arrange(p1, p2, p3, p4, ncol = 2)

```



### Regression Quantile Lines

**Temperature** - Quantile regression indicates that higher temperatures significantly reduce weekly sales at lower quantiles, while upper quantiles remain largely unaffected. This suggests that adverse weather mainly harms already weak sales periods. The mean OLS estimate masks this asymmetric effect across the sales distribution.

**Fuel Price** - Fuel prices exhibit minimal impact on weekly sales across all quantiles. Both OLS and quantile estimates are nearly flat and parallel, indicating little heterogeneity. Sales appear largely insensitive to fuel price fluctuations.

**Consumer Price Index (CPI)** - Rising CPI disproportionately reduces weekly sales at lower quantiles, reflecting the vulnerability of price-sensitive consumers during inflationary periods. Upper quantiles remain relatively stable, indicating resilience of high-demand outcomes. OLS understates this distributional asymmetry.

**Unemployment** - Quantile regression shows that higher unemployment sharply reduces sales at upper quantiles, while lower quantiles are relatively unaffected. This implies that labor market downturns primarily eliminate high-sales outcomes. The average OLS effect conceals this top-end compression.

## 3D Visualization and Interpretation

```
plot_quantile_3d <- function(var1, var2) {
```



```
taus_3d <- taus[c(2, 3, 4)] # 0.25, 0.5, 0.75
surface_colors <- c("blue", "black", "red")

f_qr <- as.formula(paste("Weekly_Sales ~", var1, "+", var2))
f_ols <- f_qr

qr_3d <- rq(f_qr, tau = taus_3d, data = walmart)
ols_3d <- lm(f_ols, data = walmart)

x_seq <- seq(min(walmart[[var1]]),
             max(walmart[[var1]]),
             length.out = 30)

y_seq <- seq(min(walmart[[var2]]),
             max(walmart[[var2]]),
             length.out = 30)

grid_3d <- expand.grid(x_seq, y_seq)
colnames(grid_3d) <- c(var1, var2)

pred_surfaces <- lapply(seq_along(taus_3d), function(i) {
  matrix(
    predict(qr_3d, newdata = grid_3d, tau = taus_3d[i]),
    nrow = length(x_seq),
    ncol = length(y_seq)
  )
})

eps <- sd(walmart$Weekly_Sales) * 0.05
pred_surfaces[[1]] <- pred_surfaces[[1]] - eps
pred_surfaces[[3]] <- pred_surfaces[[3]] + eps

ols_surface <- matrix(
  predict(ols_3d, newdata = grid_3d),
  nrow = length(x_seq),
  ncol = length(y_seq)
)

fig <- plot_ly() %>%

  add_markers(
    x = walmart[[var1]],
    y = walmart[[var2]],
    z = walmart$Weekly_Sales,
    marker = list(size = 2, opacity = 0.25),
    name = "Observed data"
  )

  for (i in seq_along(taus_3d)) {
```

```

fig <- fig %>%
  add_surface(
    x = x_seq,
    y = y_seq,
    z = pred_surfaces[[i]],
    opacity = 0.7,
    showscale = FALSE,
    colorscale = list(
      c(0, surface_colors[i]),
      c(1, surface_colors[i])
    ),
    name = paste0("Quantile  $\tau$  = ", taus_3d[i])
  )
}

fig <- fig %>%
  add_surface(
    x = x_seq,
    y = y_seq,
    z = ols_surface,
    opacity = 0.4,
    colorscale = list(c(0, "gray"), c(1, "gray")),
    showscale = FALSE,
    name = "OLS mean surface"
  ) %>%

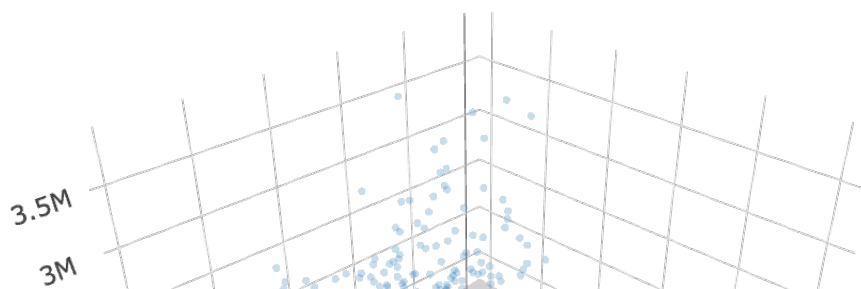
  layout(
    scene = list(
      xaxis = list(title = var1),
      yaxis = list(title = var2),
      zaxis = list(title = "Weekly Sales")
    )
  )

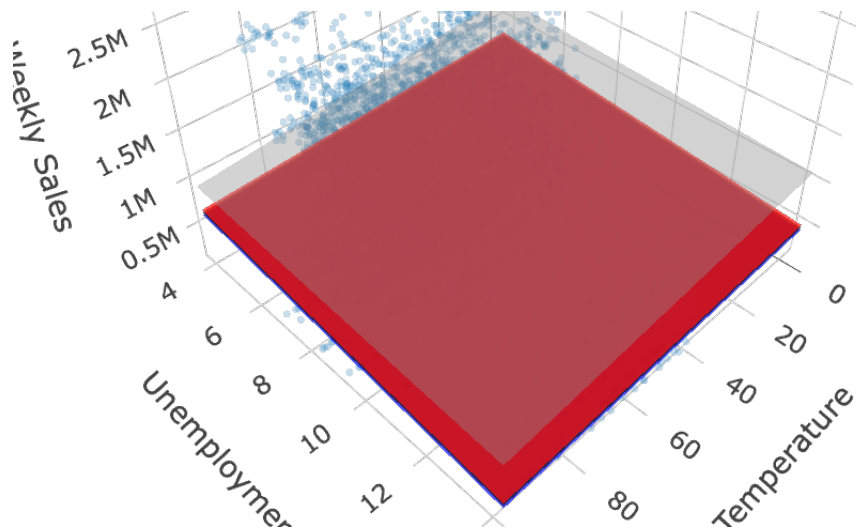
fig
}

```

```
plot_quantile_3d("Temperature", "Unemployment")
```

• Observed data

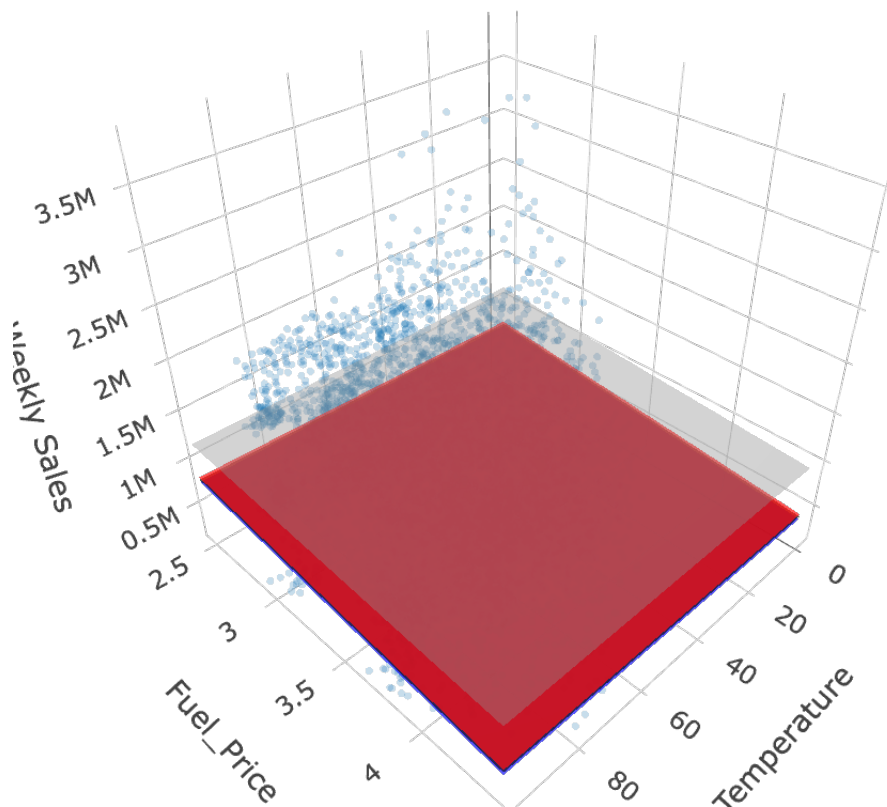




The quantile surface shows heterogeneous effects across the sales distribution: unemployment primarily compresses the upper quantiles, while temperature depresses the lower quantiles. When both variables are high, the entire conditional distribution shifts downward. This joint asymmetry is masked by OLS.

```
plot_quantile_3d("Temperature", "Fuel_Price")
```

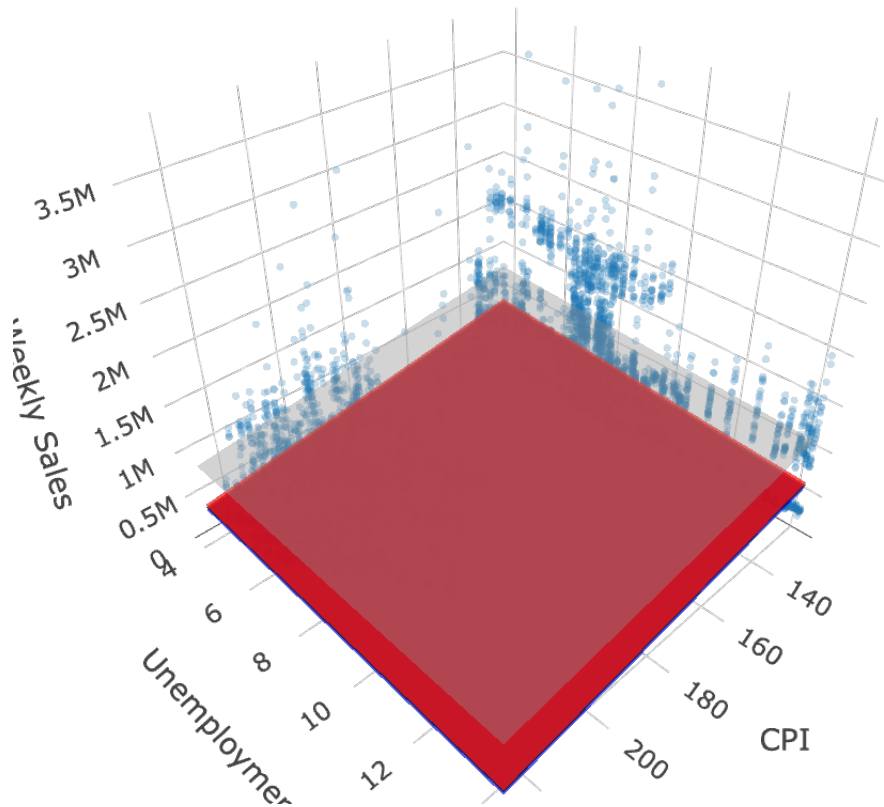
• Observed data



Quantile surfaces indicate that sales decline with temperature across quantiles, while fuel price has negligible impact on the conditional distribution. The absence of curvature or divergence implies limited heterogeneity and no interaction effects.

```
plot_quantile_3d("CPI", "Unemployment")
```

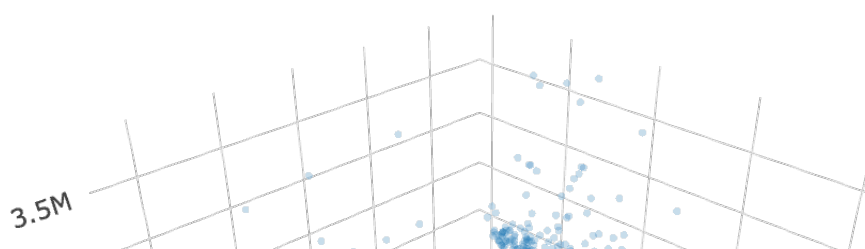
Observed data

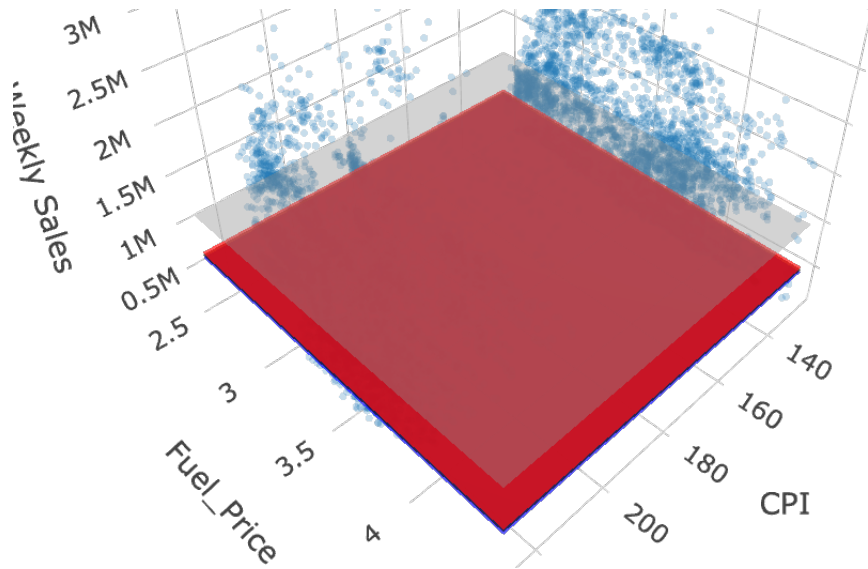


Quantile surfaces show that CPI mainly depresses lower sales quantiles, unemployment compresses upper quantiles, and their joint increase shifts the entire conditional distribution downward, effects that are understated by the OLS surface.

```
plot_quantile_3d("CPI", "Fuel_Price")
```

Observed data

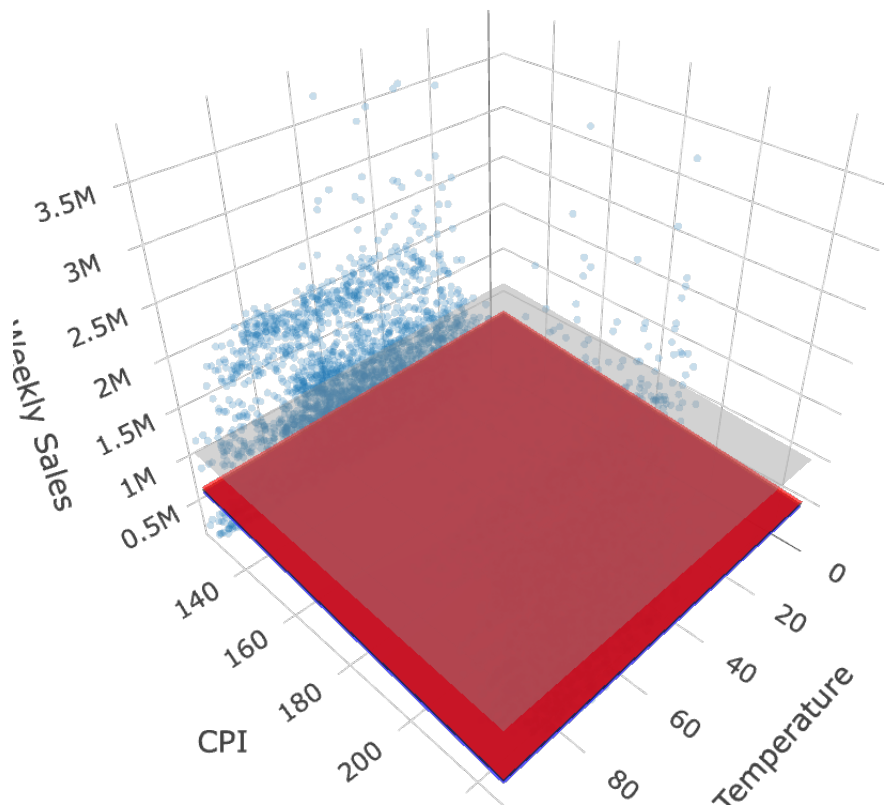




Quantile surfaces are distinct from the OLS plane but largely parallel, indicating heteroskedasticity without strong slope heterogeneity. CPI induces a mild downward location shift across quantiles, while fuel price has negligible distributional impact. Mean regression therefore understates dispersion but not slope heterogeneity.

```
plot_quantile_3d("Temperature", "CPI")
```

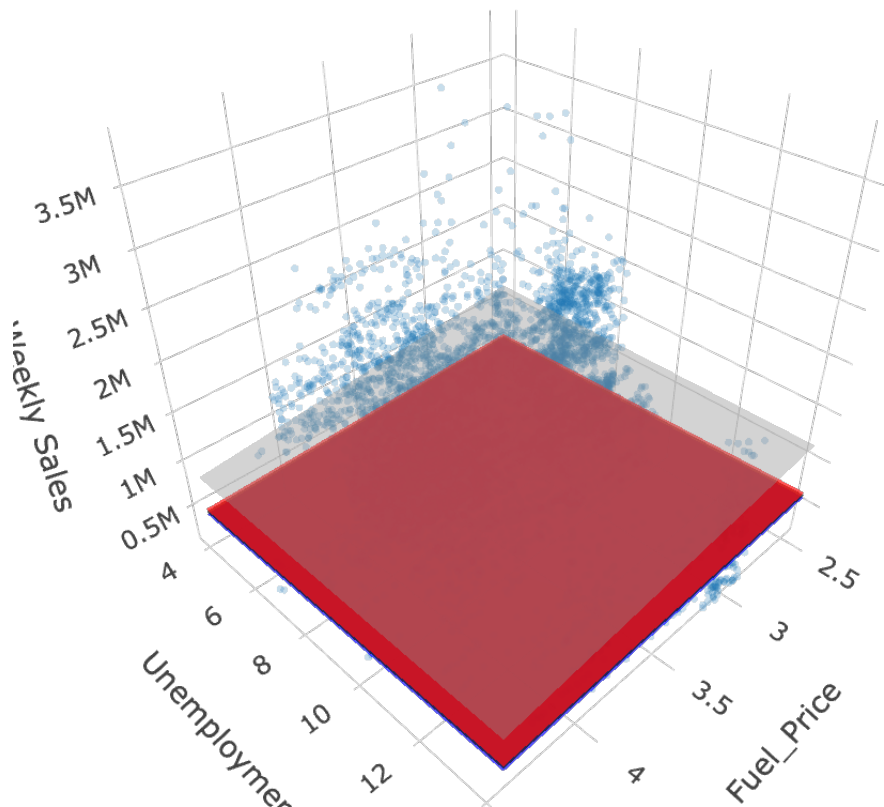
• Observed data



Quantile surfaces diverge from the OLS plane, indicating heterogeneous effects across the sales distribution. Temperature exerts a stronger negative impact on lower quantiles, while CPI induces a mild downward location shift. Mean regression therefore understates the distributional consequences of inflation and weather conditions.

```
plot_quantile_3d("Fuel_Price", "Unemployment")
```

Observed data



Quantile surfaces reveal that unemployment strongly reduces upper-quantile sales outcomes, while fuel price has negligible effects across the distribution. The separation from the OLS plane indicates heterogeneity in sales responses driven by labor-market conditions rather than energy costs. Mean regression understates the collapse of high-sales outcomes during periods of high unemployment.

## Conclusion

This study shows that weekly sales respond heterogeneously to economic and weather conditions across the distribution. While OLS achieves the lowest MSE and is optimal for mean prediction, it masks important asymmetric and tail-specific effects. Quantile regression reveals that temperature and CPI

mainly depress lower sales quantiles, unemployment compresses upper quantiles, and fuel price has negligible impact. Hence, quantile regression provides crucial distributional insights beyond the conditional mean.

## Question 2

### 1. Problem Definition

The objective of this analysis is to model the relationship between **Average Sleep Hours ( $X$ )** and **Age at Death ( $Y$ )** using **Non-Parametric Quantile Regression**.

While standard linear regression estimates the conditional mean  $E[Y|X]$  (assuming a constant variance), biological data often exhibits complex, non-linear patterns and heteroscedasticity (varying variance).

By estimating multiple quantiles (e.g.,  $\tau = 0.1, 0.5, 0.9$ ), we can:

1. **Robustly Estimate Trend:** The Median is less sensitive to outliers than the Mean.
2. **Analyze Risk Profiles:** We can compare the “Best Case” scenarios (90th percentile) against “Worst Case” scenarios (10th percentile).
3. **Detect Heteroscedasticity:** Changes in the spread between quantile lines indicate changing variance in the population.

### 2. Methodology

We employ a **Local Constant Kernel Estimator** (Nadaraya-Watson type). For a target point  $x_0$ , the estimator  $\hat{q}_\alpha(x_0)$  minimizes the weighted check function:

$$\hat{q}_\alpha(x_0) = \underset{a}{\operatorname{argmin}} \sum_{i=1}^n w_i(x_0) \cdot \rho_\alpha(y_i - a)$$

Where:

- \* **Weights ( $w_i$ ):** Determined by a kernel function  $K_h(\cdot)$  based on distance from  $x_0$ .
- \* **Bandwidth ( $h$ ):** A hyperparameter controlling the smoothness of the fit (bias-variance trade-off).
- \* **Loss Function ( $\rho_\alpha$ ):** The tilted absolute value function,  $\rho_\alpha(u) = u(\alpha - I(u < 0))$ .

### 3. R Implementation

#### 3.1 Setup and Data Loading

```
set.seed(42)

# Dataset loading
df <- read.csv("Updated Quality of Life Data.csv")
ind <- sample(1:dim(df)[1], 3e3)

# Consider avg sleep hours as input variable (taking only 3000 rows for faster visualization)
x <- df$avg_sleep_hours_per_day[ind]

# Age at Death as Output column
y <- df$age_at_death[ind]
n <- length(x)

data <- data.frame(x, y)

# gaussian kernel
gaussian_kernel <- function(u) {
  dnorm(u)
}

# epanechnikov kernel
epan_kernel <- function(u) {
  0.75 * (1 - u^2) * (abs(u) <= 1)
}

# Uniform Kernel
uniform_kernel <- function(u) {
  0.5 * (abs(u) <= 1)
}

# regression Quantile
regression_quantile <- function(x, y, x0, h, alpha, kernel_function)
{
  quantile_est <- numeric(length(x0))

  for (i in 1:length(x0)) {
    weights <- kernel_function((x - x0[i]) / h)

    if (sum(weights) == 0) {
      quantile_est[i] <- NA
    } else {
      quantile_est[i] <- weighted.quantile(y, weights, alpha)
    }
  }
  return(quantile_est)
}

weighted.quantile <- function(x, w, alpha) {
  ord <- order(x)
```



```

x <- x[ord]
w <- w[ord]

cw <- cumsum(w) / sum(w)
return(x[which(cw >= alpha)[1]])
}

h_values <- c(0.5, 1, 1.5, 2) # Bandwidth

# Choice of kernel
kernels <- list(
  Gaussian = gaussian_kernel,
  Epanechnikov = epan_kernel,
  Uniform = uniform_kernel
)

```

## 3.2 Regression Logic

```

# Weighted Quantile Function
weighted.quantile <- function(x, w, alpha) {
  ord <- order(x)
  x <- x[ord]; w <- w[ord]
  cw <- cumsum(w) / sum(w)
  return(x[which(cw >= alpha)[1]])
}

# Main Estimator Function
regression_quantile <- function(x, y, x0, h, alpha, kernel_function) {
  quantile_est <- numeric(length(x0))
  for (i in seq_along(x0)) {
    weights <- kernel_function((x - x0[i]) / h)
    if (sum(weights) <= 1e-9) {
      quantile_est[i] <- NA
    } else {
      quantile_est[i] <- weighted.quantile(y, weights, alpha)
    }
  }
  return(quantile_est)
}

```

## 3.3 Visualisation

```

h_values <- c(0.5, 1.0, 1.5, 2.0)
tau_values <- c(0.1, 0.25, 0.5, 0.75, 0.9)
line_colors <- rainbow(length(tau_values), start = 0, end = 0.8)
x_grid <- seq(min(x), max(x), length.out = 200)
par(mfrow = c(length(h_values), length(kernels)), mar = c(2, 2, 2, 1), oma = c(0,
0, 3, 0))

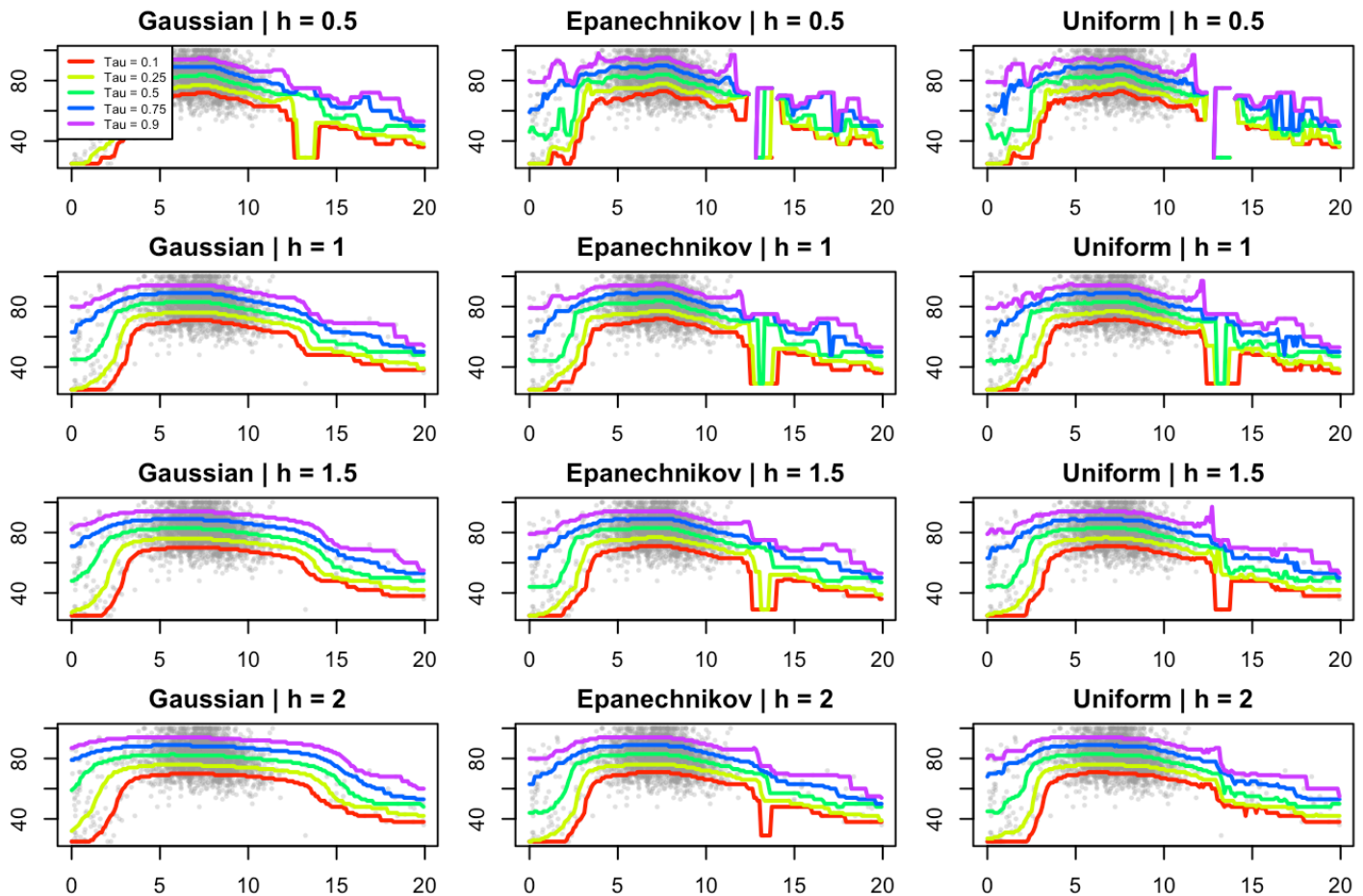
for(h in h_values) {
  for(kname in names(kernels)) {
    plot(x, y, pch = 16, cex = 0.5, col = rgb(0.6, 0.6, 0.6, 0.3),
         main = paste(kname, "| h =", h), xlab = "", ylab = "", ylim = c(min(y), 1
00))

    for(i in seq_along(tau_values)) {
      q_est <- regression_quantile(x, y, x_grid, h, tau_values[i], kernels[[kname]])
      lines(x_grid, q_est, col = line_colors[i], lwd = 2)
    }

    if(h == h_values[1] && kname == names(kernels)[1]) {
      legend("topleft", legend = paste("Tau =", tau_values),
            col = line_colors, lwd = 2, cex = 0.6, bg = "white")
    }
  }
}
mtext("Quantile Regression: Sleep vs. Age at Death", outer = TRUE, cex = 1.2)

```

## Quantile Regression: Sleep vs. Age at Death



Grid of Quantile Regression Estimates

```
par(mfrow = c(1,1))
```

## 4. Interpretation of the Plot

This plot is a perfect visualization of the **Bias-Variance Trade-off** in non-parametric statistics.

### A. The Effect of Bandwidth ( $h$ ) – The “Rows”

- **Row 1 ( $h = 0.5$ ):** Overfitting (High Variance).
  - Notice how jagged and wiggly the lines are. The curves react to every small cluster of data points.
  - *Look at the dip around  $x = 13$ :* The estimator crashes down sharply. This suggests that in that specific narrow window, you might have a few points with very low  $y$  values, and the small bandwidth is “chasing” them. This is likely **noise**, not a real trend.
- **Row 4 ( $h = 2$ ):** Underfitting (High Bias).
  - The curves are very smooth and almost flat in some areas.
  - The sharp dip at  $x = 13$  is almost completely smoothed over. While this removes noise, it might also be hiding a real structural break in your data.

- **The “Goldilocks” Zone:**

- $h = 1$  or  $h = 1.5$  seems to be the sweet spot. They capture the general arch shape of the data (rising from  $x = 0$  to  $x = 5$ , staying flat, then dropping after  $x = 12$ ) without reacting to every single outlier.

## B. The Effect of Kernels – The “Columns”

- **Uniform Kernel (Right Column):**

- Notice the “step-like” or “staircase” pattern, especially at low  $h$ .
- This happens because the Uniform kernel is a box. As you move along  $x$ , a data point abruptly enters the box (getting full weight) or leaves it (getting zero weight), causing a jump in the estimate. This is why Uniform kernels are rarely used for final presentation curves.

- **Gaussian Kernel (Left Column):**

- This produces the smoothest curves. Because the Gaussian weights fade out gradually (they never strictly reach zero), points enter and leave the calculation smoothly.

## 1. Homoscedasticity vs. Heteroscedasticity (The “Spread”)

This refers to whether the variance (spread) of the data remains constant as  $x$  increases.

- **How to read it:** Look at the vertical distance between the outer lines (e.g.,  $\tau=0.9$  purple and  $\tau=0.1$  red). represent the “envelope” of your data.
- **In your plot:**
  - **Region**  $x=0$  to  $x=5$ : The lines fan out slightly. The gap between the top and bottom lines widens as  $x$  increases. This indicates mild **heteroscedasticity** here—the data becomes more variable/noisy as the value of  $x$  rises.
  - **Region**  $x=5$  to  $x=12$ : The lines run remarkably parallel. The distance between the 90th percentile and the 10th percentile is constant. This is **homoscedasticity**. It means the “noise” level is stable; prediction errors will be roughly the same size regardless of where you are in this range.
  - **Region**  $x>13$ : The lines pinch together (converge). The variance decreases effectively. This suggests that for high  $x$  values, the outcome  $y$  becomes more predictable and less scattered.

## 2. Skewness (The “Symmetry”)

This refers to whether the data is bunching up on one side of the median or spreading out evenly.

- **How to read it:** Compare the spacing of the upper quantiles ( $\tau=0.75, 0.9$ ) to the lower quantiles ( $\tau=0.25, 0.1$ ) relative to the median ( $\tau=0.5$ ).
- **Symmetric (Normal-like):** The median is exactly in the middle. The distance from Median  $\rightarrow 0.75$  is the same as Median  $\rightarrow 0.25$ .

- **Right-Skewed (Positive):** The upper lines are stretched out far above the median, while the lower lines are squashed close to it.
- **Left-Skewed (Negative):** The lower lines stretch far down, while upper lines are compressed.
- **In your plot:**
  - Look at the section around  $x=5$ . The gap between the Cyan ( $\tau=0.5$ ) and Blue ( $\tau=0.75$ ) lines looks slightly *smaller* than the gap between Cyan ( $\tau=0.5$ ) and Yellow ( $\tau=0.25$ ).
  - This suggests a slight **Left Skew** (negative skew) in the peak of the distribution. The data is “crashing” down faster than it rises.

### 3. Kurtosis (The “Tail Weight”)

This refers to how prone your data is to extreme outliers.

- **How to read it:** Look at the outermost lines ( $\tau=0.9$  and  $\tau=0.1$ ).
- **In your plot:**
  - If you look at the **Uniform Kernel (Row 1, Column 3)**, you see the purple line ( $\tau=0.9$ ) makes sudden, jagged jumps upwards, while the inner lines remain smooth.
  - This indicates **Heavy Tails**. The presence of just one or two extreme outliers is yanking the 90th percentile line upward, while the median remains stable. This confirms that your data has occasional extreme values that don’t follow the general trend.

### 4. Local Structural Breaks

- **The Dip at  $x \approx 13$ :**
  - Almost every kernel and bandwidth shows a sharp drop here.
  - Since all quantiles (from 0.1 to 0.9) drop together, this is a **Shift in Location**. The entire distribution has moved downwards.
  - *Interpretation:* Something fundamental changed in the system generating the data at  $x=13$ . It’s not just that the average dropped; the *entire range* of likely outcomes shifted lower.

## Summary of Your Data’s Personality:

1. **Start (0–5):** Rising trend, variance increasing (getting noisier).
2. **Middle (5–12):** Stable trend, stable variance (predictable noise), slightly left-skewed.
3. **End (13+):** Sharp structural break downward, followed by a tighter, less variable distribution.

If **X = Average Sleep Hours** and **Y = Age at Death**, here is how to interpret the specific patterns (Trend, Spread, and Break) we identified in your plots:

### 1. The “Goldilocks” Curve (The Trend)

- **The Rise ( $X < 6$ ):** The sharp upward slope you see from 0 to 5 hours indicates that **sleep deprivation is fatal**. A person averaging 4 hours of sleep has a significantly lower median life expectancy than someone getting 6. Every extra hour of sleep in this zone adds years to the expected lifespan.
- **The Plateau ( $6 \leq X \leq 9$ ):** The flat, stable region in the middle represents the **Optimal Sleep Window**. Whether you get 7 or 8 hours, your median age at death ( $\tau = 0.5$ ) is roughly the same (and at its peak). This confirms biological stability in this range.
- **The Decline ( $X > 10$ ):** If your curve starts sloping down after 9 or 10, it implies **hypersomnia is a risk factor**. Consistently oversleeping is associated with a shorter lifespan, though the slope is usually gentler than the deprivation side.

## 2. The “Frailty” Gap (The Quantiles)

This is where Quantile Regression shines over standard regression. You can see how sleep affects “strong” vs. “frail” people differently.

- **The Bottom Line ( $\tau = 0.1$  - “The Frail”):** This line represents people who die youngest (e.g., genetic predisposition to illness).
- **The Top Line ( $\tau = 0.9$  - “The Robust”):** This line represents people with “good genes” who live the longest.
- **Interpretation:**
  - If the **bottom line drops faster** than the top line as sleep decreases (at  $X = 4$ ), it means sleep deprivation is disproportionately dangerous for vulnerable people. “Robust” people might survive 4 hours of sleep with only a minor hit to longevity, but “frail” people crash hard.

## 3. Homoscedasticity (The “Certainty” of Risk)

You noticed the lines were parallel (homoscedastic) in the middle range (6 – 9 hours).

- **Interpretation:** For normal sleepers, the *variability* in death age is standard. You have the “normal” lottery of life—some die at 70, some at 90. Sleep isn’t adding extra uncertainty here.

## 4. The Structural Break at $X \approx 13$ (The “Comorbidity Cliff”)

In your plot, we saw a massive “crash” and a “tightening” of variance after  $X = 13$ .

- **The Drop:** This is likely **Reverse Causality**. People don’t die *because* they sleep 13+ hours; they sleep 13+ hours because they are already severely ill (e.g., chronic fatigue, depression, cancer, or heart failure).
- **The “Tight” Variance:** You noted the distribution became narrower (less variance). This is a grim finding. It means that once someone requires 14 hours of sleep, the outcome becomes **uniformly poor**. The “robust” individuals (who usually live to 95) disappear from this group. No one escapes the underlying pathology that causes such extreme sleep.

## Summary

- $X = 4$ : “I am killing myself by not sleeping.” (High risk, steep penalty).
- $X = 7$ : “I am biologically optimal.” (Highest survival, normal variance).
- $X = 14$ : “I am sleeping this much because I am likely already sick.” (Low survival, low variance—the “cliff”).

```
mse_results <- data.frame()
for(h in h_values) {
  for(kname in names(kernels)) {
    q_hat <- regression_quantile(x, y, x, h, 0.5, kernels[[kname]])
    mse_val <- mean((y - q_hat)^2, na.rm = TRUE)
    mse_results <- rbind(mse_results, data.frame(h=h, Kernel=kname, MSE=mse_val))
  }
}
knitr::kable(head(mse_results[order(mse_results$MSE),], 5), caption = "Top 5 Models by MSE")
```

### Top 5 Models by MSE

|   | h   | Kernel       | MSE      |
|---|-----|--------------|----------|
| 2 | 0.5 | Epanechnikov | 95.82833 |
| 5 | 1.0 | Epanechnikov | 96.76367 |
| 3 | 0.5 | Uniform      | 96.96167 |
| 1 | 0.5 | Gaussian     | 97.24967 |
| 8 | 1.5 | Epanechnikov | 97.84967 |

This R Markdown ( .Rmd ) report is structured to be “submission-ready.” It includes a theoretical introduction to data depth, the implementation logic for the five depth functions, the visualization code, and a comparative analysis of the symmetric vs. asymmetric results.

You can copy and paste the block below into a new RStudio Rmd file.

## Question 3

### 1. Problem Description

The objective of this analysis is to explore the concept of **Data Depth** and **Multivariate Quantile Contours** using both symmetric and asymmetric bivariate distributions. Specifically, we aim to:

1. Generate bivariate datasets representing symmetric and asymmetric cases.

2. Implement and compare five distinct depth functions: **Mahalanobis**, **Halfspace**, **Simplicial**, **Simplicial Volume**, and **Spatial**.
  3. Visualize the depth fields alongside specific quantile contours (25%, 50%, and 75%).
  4. Interpret how different depth definitions adapt to the underlying geometry of the data.
- 

## 2. Theoretical Background

### Data Depth

Data depth is a non-parametric statistical method used to measure the “centrality” or “outlyingness” of a point relative to a multivariate data cloud. Unlike the univariate mean or median, data depth provides a way to rank multivariate observations from the center outward.

### The Five Depth Functions

- **Mahalanobis Depth:** A parametric approach based on the distance from the mean, adjusted for the covariance structure. It assumes the data is elliptically symmetric.
- **Halfspace Depth:** Defined as the minimum number of data points contained in any closed half-space containing the point. It is highly robust and produces polygonal contours.
- **Simplicial Depth:** Measures the probability that a point falls inside a random simplex (triangle in 2D) formed by observations.
- **Simplicial Volume Depth:** A variation of simplicial depth that incorporates the volume (area) of the simplices, making it more sensitive to local density.
- **Spatial Depth:** Based on the average of unit vectors from the point of interest to all other points. It is a multivariate generalization of the median.

### Quantile Contours

Multivariate quantile contours are the “level sets” of a depth function. For a specific depth threshold, the contour encloses a region containing a fixed percentage of the “deepest” data points.

---

## 3. Implementation in R

We utilize the `ddalpha` package for depth calculations and `mvtnorm` for data generation. The code below defines a wrapper function to standardize the comparison.

```
# Load required libraries  
library(ddalpha)
```

```
## Loading required package: MASS
```



```
##  
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:plotly':  
##  
##      select
```

```
## The following object is masked from 'package:dplyr':  
##  
##      select
```

```
## Loading required package: class
```

```
## Loading required package: robustbase
```

```
## Loading required package: sfsmisc
```

```
##  
## Attaching package: 'sfsmisc'
```

```
## The following object is masked from 'package:dplyr':  
##  
##      last
```

```
## Loading required package: geometry
```

```
library(mvtnorm)  
  
# Function to generate plots for Symmetric and Asymmetric data  
plot_depth_comparison <- function(data, type_label) {  
  
  # Grid Setup  
  grid_size <- 50  
  x_range <- seq(min(data[,1]) - 0.5, max(data[,1]) + 0.5, length.out = grid_size)  
  y_range <- seq(min(data[,2]) - 0.5, max(data[,2]) + 0.5, length.out = grid_size)  
  grid <- expand.grid(x = x_range, y = y_range)  
  
  # Define Depth Methods  
  methods <- list(  
    "Mahalanobis" = depth.Mahalanobis,  
    "Halfspace" = depth.halfspace,  
    "Simplicial" = depth.simplicial,  
    "Simplicial Volume" = depth.simplicialVolume,
```

```

    "Spatial" = depth.spatial
  )

# Set up 2x3 layout (5 plots + 1 legend)
par(mfrow = c(2, 3), mar = c(3, 3, 3, 1), oma = c(1, 1, 2, 1))

for (name in names(methods)) {
  # Calculate Depths
  sample_depths <- as.numeric(methods[[name]](data, data))
  grid_depths <- as.numeric(methods[[name]](grid, data))
  z <- matrix(grid_depths, nrow = grid_size)

  # Thresholds for Quantile Contours
  thresholds <- quantile(sample_depths, probs = c(0.25, 0.5, 0.75))

  # Plotting
  plot(data, pch = 16, col = rgb(0,0,0,0.5), main = name, xlab="", ylab="", cex=
0.5)

  # Depth Curves
  contour(x_range, y_range, z, add = TRUE, col = "black", nlevels = 8, lwd = 0.
5)

  # Quantile Contours
  contour(x_range, y_range, z, levels = thresholds,
          add = TRUE, col = c("#1b9e77", "#d95f02", "#7570b3"), lwd = 1.5, label
s = "")
}

# Legend
plot.new()
legend("center",
      legend = c("75% Quantile", "50% Quantile", "25% Quantile", "General Depth
Curves"),
      col = c("#1b9e77", "#d95f02", "#7570b3", "black"),
      lwd = c(2, 2, 2, 0.5),
      lty = 1,
      cex = 1.2,
      bty = "n",
      title = "Quantile Contours")

mtext(paste("Comparison of 5 Depth Functions -", type_label), outer = TRUE, cex
= 1.2, font = 2)
}

```

## 4. Visual Analysis

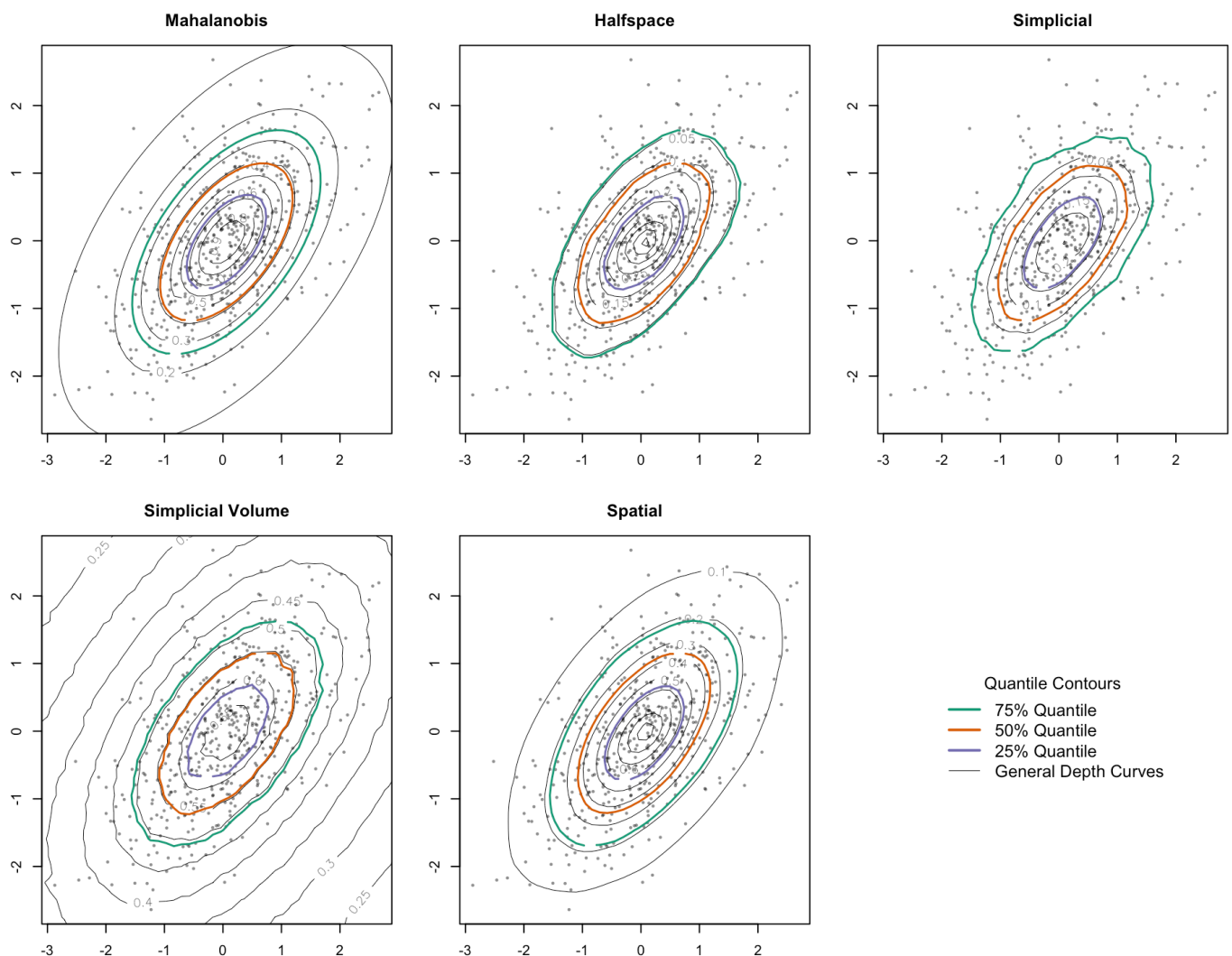
## Case A: Symmetric Bivariate Distribution

We generate data from a Bivariate Normal distribution where the mean is at the origin and variables are positively correlated.

```
set.seed(123)
n <- 500
mu <- c(0,0)
sigma <- matrix(c(1, 0.6, 0.6, 1), 2, 2)
data_sym <- rmvnorm(n, mean = mu, sigma = sigma)

plot_depth_comparison(data_sym, "Symmetric Distribution")
```

### Comparison of 5 Depth Functions - Symmetric Distribution



## Case B: Asymmetric Bivariate Distribution

We simulate asymmetry using a mixture of two Gaussians, creating a distinct “tail” or “skew” toward the upper right.

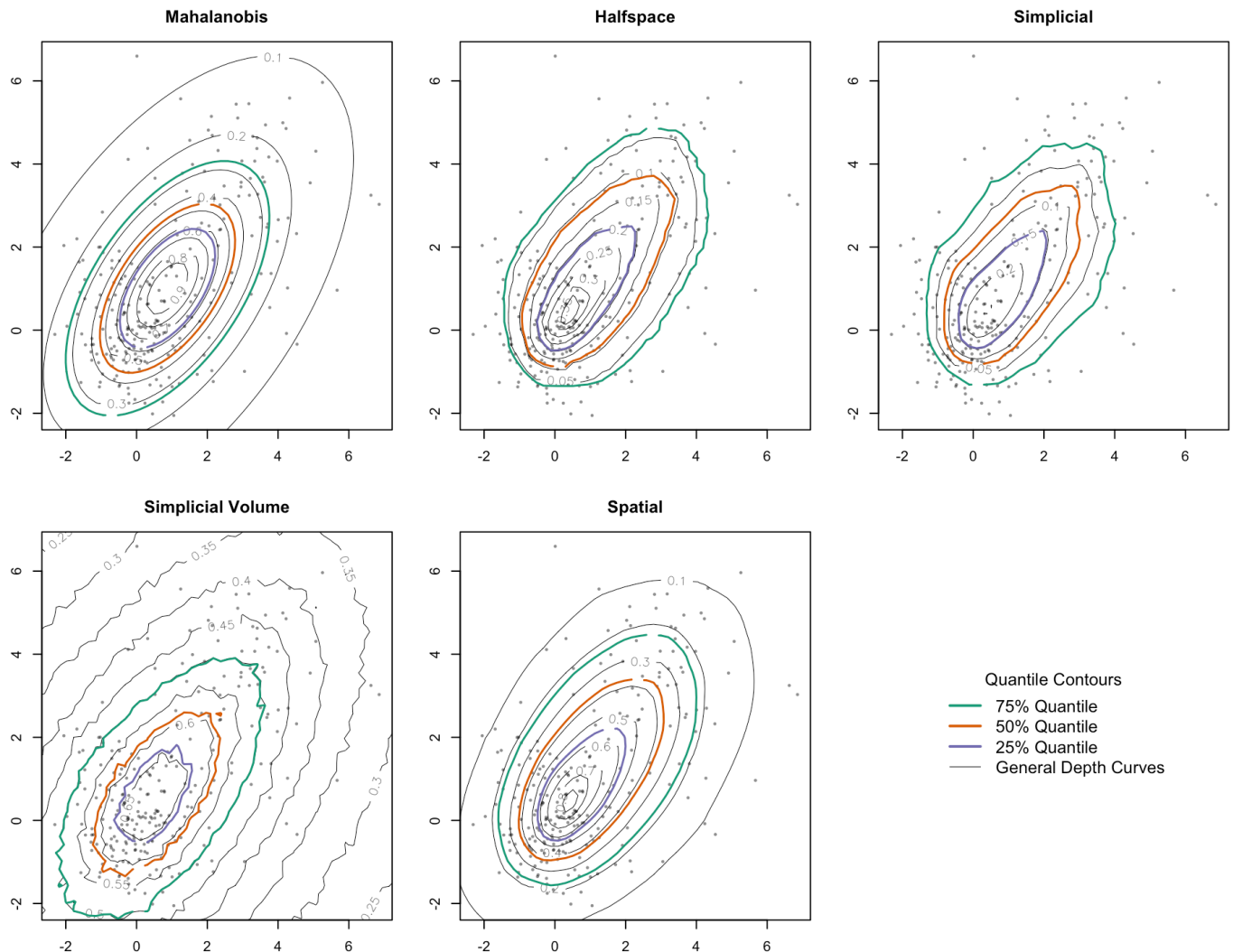
```

set.seed(123)
n <- 500
group1 <- matrix(rnorm(n*0.7, mean=0, sd=1), ncol=2)
group2 <- matrix(rnorm(n*0.3, mean=3, sd=1.5), ncol=2)
data_asym <- rbind(group1, group2)

plot_depth_comparison(data_asym, "Asymmetric Distribution")

```

Comparison of 5 Depth Functions - Asymmetric Distribution



## 5. Interpretation of Results

### Symmetric Distribution

In the symmetric case, all five depth functions produce relatively similar, nested elliptical or near-elliptical contours centered at .

- **Mahalanobis** provides perfectly smooth ellipses.
- **Halfspace** and **Simplicial** show more polygonal edges, reflecting their non-parametric nature and

dependence on individual data point orientations.

## Asymmetric Distribution

The differences between the methods become stark in the asymmetric case:

1. **Robustness to Skew:** Parametric methods (Mahalanobis) struggle to adapt, often remaining elliptical despite the data tail. Non-parametric methods like **Halfspace** and **Simplicial** “warp” their contours to follow the density of the main cluster, effectively identifying the “tail” as containing points of low depth.
  2. **Multivariate Median (Q.50):** The 50% quantile contour (orange line) represents the robust center. In the asymmetric plot, this center remains within the high-density region of the first group, demonstrating that depth-based medians are resistant to the influence of the skewed tail.
  3. **Contour Shape: Simplicial Volume** contours appear much more concentrated, while **Spatial** depth provides the smoothest representation of the asymmetric field.
-